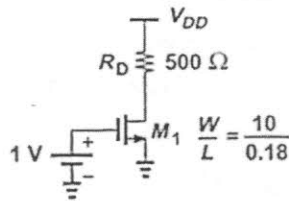


6. In the circuit shown in Fig.6, the minimum allowable value of V_{DD} if M1 must not enter the triode region is _____. Assume $V_{TH}=0.4V$ and $\mu_n C_{ox} = 100 \mu A/V^2$



- A) 1.6 V
B) 1.1 V
C) 1 V
D) 0.6 V

Fig. Q6

7. In a CS amplifier with degeneration, if $g_m=1mS$, $R_D = 5k\Omega$, $R_s=1k\Omega$, gain is _____
A) 2.5 B) 5
C) -2.5 D) -5
8. In amplifier circuits, the MOSFETs are operated in _____ region.
A) cutoff B) saturation
C) Any D) Linear or active
9. If common source amplifier with MOSFETs M1 and M2, where input is applied to M1, then voltage gain of it is _____
A) $-g_{m1}/g_{m2}$ B) $+g_{m1}/g_{m2}$
C) $-g_{m2}/g_{m1}$ D) $+g_{m2}/g_{m1}$
10. The differential amplifier using MOSFETs requires _____ input devices
A) One B) two
C) More than two D) three
11. The CMRR of differential amplifier should be _____ for better noise rejection.
A) anything B) moderate
C) small D) large
12. Active current mirror is also known as _____ amplifier.
A) Operational transimpedance B) conducting
C) Operational transconductance D) None of these
13. The device model used to find the transfer function of amplifier circuits is _____
A) Low frequency model B) High frequency model
C) Large signal model D) Hybrid model
14. Which theorem is useful in determining the frequency response of amplifiers?
A) Miller's theorem B) Reciprocity theorem
C) Millman's theorem D) Superposition theorem
15. The gain of the amplifier at the cutoff frequency of the device is _____
A) 1 dB B) 100 dB
C) 0 dB D) 10 dB
16. The number of poles in the system _____ with cascade connection of amplifiers.
A) Do not change B) decreases
C) increases D) Reduce to half
17. In current shunt feedback the input impedance _____
A) increases B) decreases
C) may increase or decrease D) None of these
18. The type of feedback used in oscillator circuit is _____
A) negative B) positive
C) negative and positive D) negative or positive
19. The main features of a large-signal amplifier is the circuit's _____.
A) Power efficiency B) Maximum power limitation
C) Impedance matching to the output device D) All the above
20. Which type of power amplifier is biased for operation at less than 180° of the cycle?
A) Class A B) Class B
C) Class AB D) Class C

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

Marks BT* CO* PO*

1. a) Draw the circuit diagram of a CE stage with resistive divider bias circuit and without emitter degeneration. Calculate the collector current of the transistor if $R_1=17\text{ k}\Omega$, $R_2=8\text{ k}\Omega$ (connected between base and the ground), $R_C=5\text{ k}\Omega$, $V_{CC}=2.5\text{ V}$, $I_S=10^{-17}\text{ A}$ and $\beta=100$. Verify that transistor operates in active mode.

8	L*3	1	1
---	-----	---	---
- b) For the circuit shown in **Fig.Q1(b)**, draw the small signal circuit and derive the expression for voltage gain, input and output impedance. Assume $\lambda=0$.

8	L3	2	1
---	----	---	---
2. a) In the circuit shown in **Fig. Q2(a)** BJT operates at the edge of active region. The collector bias voltage is 0.85 V , $R_B=100\text{ k}\Omega$, $R_C=1\text{ k}\Omega$ and $V_{CC}=2.5\text{ V}$. Calculate collector current and β of the transistor.

8	L3	1	1
---	----	---	---
- b) Show that the voltage gain of source follower is less than one.

8	L2	2	1
---	----	---	---

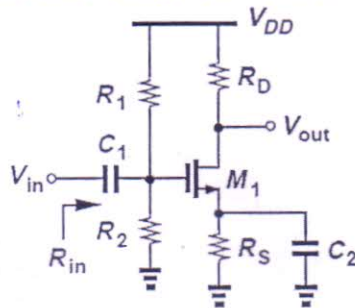


Fig.Q1(b)

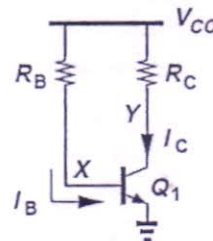


Fig.Q2(a)

3. a) Determine the expression for voltage gain of the circuit shown in **Fig. Q3(a)**. Use small signal equivalent circuit. Assume $V_A = \infty$.

8	L2	1	1
---	----	---	---
- b) Calculate the small-signal voltage gain of the CS stage shown in **Fig. 3(b)** if $I_D = 1\text{ mA}$, $\mu_n C_{ox} = 100\mu\text{A/V}^2$, $V_{th} = 0.5\text{ V}$ and $\lambda=0$. Verify that M1 operates in saturation.

8	L3	2	1
---	----	---	---

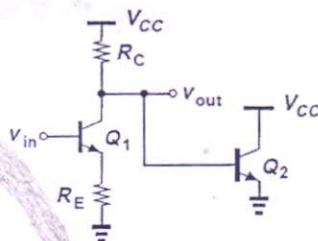


Fig. Q3(a)

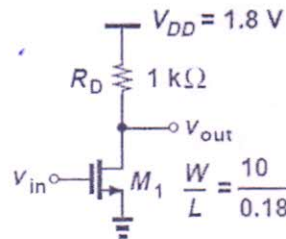


Fig. Q3(b)

Unit – II

4. a) Derive the expression for I_{copy} and calculate the value of I_{REF} , if $I_{copy} = 0.4\mu\text{A}$, in the circuit shown in **Fig.Q4(a)**.

8	L3	3	1
---	----	---	---
- b) Derive the expression for input impedance of the circuit shown in **Fig.Q4(b)**.

8	L2	4	1
---	----	---	---

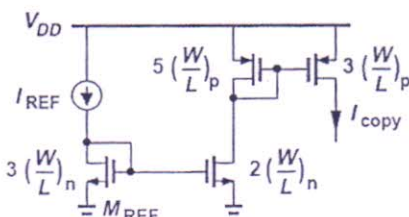


Fig.Q4(a)

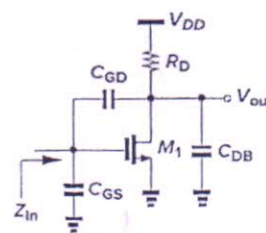


Fig.Q4(b)

5. a) Sketch the circuit of active current mirror and explain its working. 8 L2 3 1
 b) Derive the expression for transfer function of common source amplifier shown in Fig. Q5(b). Use high frequency model of the MOSFET.

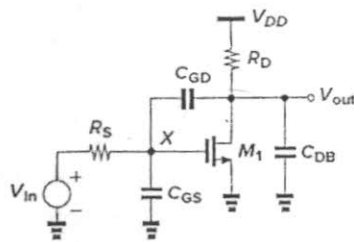


Fig. Q5(b)

6. a) Derive the expression for small signal voltage gain of source coupled differential amplifier with resistive load. 8 L2 4 1
 b) Derive the expression for transfer function of source follower circuit shown in Fig.Q6(b). Use high frequency model of the MOSFET.

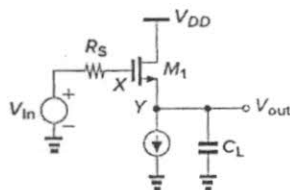


Fig. Q6(b)

- Unit – III
7. a) Prove that voltage series feedback increases the input impedance and decreases the output impedance. 8 L2 5 1
 b) Derive the expression for ac power, dc power of class B power amplifier and show that maximum power conversion efficiency is 78.5%. 8 L2 5 1
8. a) For the circuit shown in Fig.Q8(a), identify the type of feedback and prove it. Also, determine gain with feedback and the feedback factor.

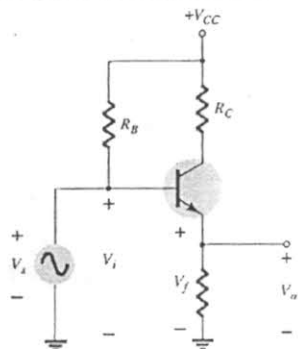


Fig. Q8(a)

- b) Derive the expression for ac power, dc power and maximum power conversion efficiency of series fed class A power amplifier. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
